Econometrica Supplementary Material

SUPPLEMENT TO "IDENTIFYING RATIONAL TYPES IN UNKNOWN ENVIRONMENTS" APPENDIX 3: FORMAL PROOFS AND LEAN VERIFICATION

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This supplement presents in full detail the formal statements and proofs supporting the identification argument in the main paper. It is organized in eight sections that mirror the logical structure of the model: stochastic trends (Section S1), public information and Bayesian learning (Section S2), the stochastic implementation mechanism (Section S3), asymptotic type identification (Section S4), competitive risks and operational pressure (Section S5), policy implications (Section S6), equilibrium implementability (Section S7), and the abstract Kakutani identification criterion (Section S8), plus machine-checked verification (Section S9). Analytical proofs are self-contained in this document; algebraic steps verified in Lean 4 are reproduced from the `Appendix_3/` replication package (Section S10). Classical measure-theoretic inputs (Kakutani, Blackwell–Dubins, fixed-point existence) remain proved here and are not rederived inside the proof assistant.

S1. RESULTS FROM THE MAIN PAPER

The following result is imported from *Identifying Rational Types in Unknown Environments: Evidence from Kidnapping in Colombia*. It operates in the time-series domain and is independent of the game-theoretic blocks below. Lemma S1.1 supplies the integration-order step.

LEMMA S1.1—$I(0)$ closed under linear combinations: *If $X_t \sim I(0)$ and $Y_t \sim I(0)$, then $aX_t + bY_t \sim I(0)$ for all $a, b \in \mathbb{R}$.*

PROOF: Covariance stationarity is preserved under finite linear combinations: for all s, t and all $a, b \in \mathbb{R}$,

$$\mathbb{E}\big[(aX_{t+s} + bY_{t+s}) - (aX_t + bY_t)\big] = 0,$$

and the autocovariance function of $aX_t + bY_t$ depends on lags only through $t - s$. Hence $aX_t + bY_t$ is integrated of order zero. *Q.E.D.*

COROLLARY S1.2—Equivalence of stochastic trends: *Let $\{y_t\}$ be a time series integrated of order one, $I(1)$. Suppose the existence of two valid decompositions:*

$$y_t = P_t^{(1)} + u_t^{(1)} = P_t^{(2)} + u_t^{(2)},$$

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where $P_t^{(1)}$ and $P_t^{(2)}$ are $I(1)$ processes and $u_t^{(1)}$, $u_t^{(2)}$ are stationary $I(0)$ processes. Then:

$$P_t^{(1)} - P_t^{(2)} \sim I(0).$$

Consequently, all valid stochastic trends derived from the same observed series are equivalent up to a stationary component.

PROOF: From the two decompositions, $P_t^{(1)} - P_t^{(2)} = u_t^{(2)} - u_t^{(1)}$. Since $u_t^{(1)}$ and $u_t^{(2)}$ are $I(0)$, Lemma S1.1 with $a = 1$ and $b = -1$ yields $P_t^{(1)} - P_t^{(2)} \sim I(0)$. Any pair of valid stochastic trends extracted from the same $I(1)$ series therefore differs by a transitory $I(0)$ component. *Q.E.D.*

S2. PUBLIC INFORMATION, LIKELIHOOD, AND BAYESIAN LEARNING

The three definitions below construct the public filtration, the likelihood kernel, and the belief process. The stability lemma closes the block.

DEFINITION S2.1—Public signal and filtration: **(i) Public signal ζ_t.** The public signal ζ_t is the random vector that summarizes what is observable at the close of period t:

$$\zeta_t = (\alpha_t, \gamma_t, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, m_t, d_t) \in \mathcal{Z}_t \subseteq \mathbb{R}^2 \times \mathcal{A}^F \times \mathcal{A}^K \times \mathcal{A}^{S,\text{disc}} \times \mathcal{M}_{\text{out}} \times \{0,1\}.$$

(ii) Public filtration $\{\mathcal{F}_t^{\text{pub}}\}$. Set $\mathcal{F}_0^{\text{pub}} := \sigma(z, \theta_V)$. For $t \geq 1$,

$$\mathcal{F}_t^{\text{pub}} := \sigma(z, \zeta_0, \ldots, \zeta_{t-1}),$$

that is, the σ-algebra generated by the initial location and by the signals observed strictly before period t begins.
(iii) Measurability. Each $\zeta_t : (\Omega, \mathcal{F}) \to (\mathcal{Z}_t, \mathcal{B}(\mathcal{Z}_t))$ is measurable, and $\sigma(z, \zeta_0, \ldots, \zeta_{t-1})$ is well defined as a sub-σ-algebra of $\mathcal{F}$.

□

DEFINITION S2.2—Predictive likelihood under a counterfactual type: Fixing a PBE with profile (χ, s^F, s^K) and a counterfactual type $\theta \in \Theta_K$, the **predictive likelihood** of period t is

$$\ell_t(\theta; J) := \mathbb{P}_{C,\theta}\big(\zeta_t \in J \mid \mathcal{F}_t^{\text{pub}}\big), \qquad J \in \mathcal{B}(\mathcal{Z}_t). \tag{S2.1}$$

The mapping $J \mapsto \ell_t(\theta; J)$ is a **probability kernel** that carries the information state $\mathcal{F}_t^{\text{pub}}$ to the law of ζ_t under type θ.

□

DEFINITION S2.3—Beliefs and PBE-consistent Bayesian updating: Let $\mu_0 \in \Delta(\Theta_K)$ be the **common prior** with full support, $\mu_0(\theta \mid z, \theta_V) > 0$ for all $\theta \in \Theta_K$. For $t \geq 1$, let $\mu_t \in \Delta(\Theta_K)$ be the distribution obtained by iterated Bayes from μ_0, using the realizations of $(\zeta_0, \ldots, \zeta_{t-1})$ and the predictive kernels $\ell_s(\theta'; J)$ induced by the same PBE.

The process $\{\mu_t\}_{t\geq 0}$ is **consistent with the PBE** if, under the ex-ante law $\mathbb{P}_\mathrm{E}$,

$$\mu_t(\theta) = \mathbb{P}_\mathrm{E}\big(\theta_K = \theta \mid \mathcal{F}_t^{\mathrm{pub}}\big) \qquad \forall\, \theta \in \Theta_K,\ \forall\, t \geq 1, \tag{S2.2}$$

interpreted $\mathbb{P}_\mathrm{E}$-almost surely upon fixing regular versions of the conditional probabilities.

□

LEMMA S2.4—Asymptotic stability of beliefs. Convergence theorem of **?**, **?**: *Under the ex-ante law $\mathbb{P}_\mathrm{E}$ of Definition S2.3, the process $\{\mu_t(\theta)\}_{t\geq 0}$ is a* ***bounded martingale*** *that converges $\mathbb{P}_\mathrm{E}$-almost surely to a limit random variable $Z_\theta \in [0,1]$ $\mathbb{P}_\mathrm{E}$-a.s.*

PROOF: The proof verifies the conditions of the convergence theorem of **???**.

1. Martingale property. Let $M = \mathbf{1}\{\theta_K = \theta\}$. By equilibrium consistency (Definition S2.3),

$$\mu_t(\theta) = \mathbb{E}^{\mathbb{P}_\mathrm{E}}[M \mid \mathcal{F}_t^{\mathrm{pub}}] \quad \mathbb{P}_\mathrm{E}\text{-a.s.}$$

As $\mathcal{F}_t^{\mathrm{pub}} \subseteq \mathcal{F}_{t+1}^{\mathrm{pub}}$, the **law of iterated expectations** yields

$$\mathbb{E}^{\mathbb{P}_\mathrm{E}}[\mu_{t+1}(\theta) \mid \mathcal{F}_t^{\mathrm{pub}}] = \mathbb{E}^{\mathbb{P}_\mathrm{E}}\big[\mathbb{E}^{\mathbb{P}_\mathrm{E}}[M \mid \mathcal{F}_{t+1}^{\mathrm{pub}}] \mid \mathcal{F}_t^{\mathrm{pub}}\big] = \mathbb{E}^{\mathbb{P}_\mathrm{E}}[M \mid \mathcal{F}_t^{\mathrm{pub}}] = \mu_t(\theta) \quad \mathbb{P}_\mathrm{E}\text{-a.s.}$$

The process is a martingale under $\mathbb{P}_\mathrm{E}$.

2. Pointwise and uniform L^1 bound. As $\mu_t(\theta)$ is a probability mass, $0 \leq \mu_t(\theta) \leq 1$ for all $t \geq 0$ (outside a $\mathbb{P}_\mathrm{E}$-null set). In particular, $\sup_{t\geq 0} \mathbb{E}^{\mathbb{P}_\mathrm{E}}[|\mu_t(\theta)|] \leq 1$, which provides the uniform L^1 bound required by Doob's theorem.

3. Almost sure convergence. Once the martingale property and the uniform L^1 bound are verified, **Doob's convergence theorem** guarantees the existence of Z_θ such that

$$\lim_{t\to\infty} \mu_t(\theta) = Z_\theta \quad \mathbb{P}_\mathrm{E}\text{-a.s.}$$

The pointwise bound $0 \leq \mu_t(\theta) \leq 1$ implies $Z_\theta \in [0,1]$ $\mathbb{P}_\mathrm{E}$-a.s. *Q.E.D.*

S3. STOCHASTIC IMPLEMENTATION MECHANISM (MDG)

This section establishes the fundamental definitions of the stochastic implementation layer (MDG), which transforms strategic intentions into materialized actions, and of the law governing the physical outcome that determines the observable results of the episode. The episode horizon is the endogenous stopping time

$$\tau := \inf\{t \geq 0 : m_t \in \mathcal{M}_{\mathrm{term}}\}, \tag{S3.1}$$

where m_t is the physical outcome, $\mathcal{M}_{\mathrm{cont}} = \{\mathrm{cont}\}$, $\mathcal{M}_{\mathrm{term}} = \{\mathrm{pay}, \mathrm{res}, \mathrm{rel}, \mathrm{kill}\}$, and $\mathcal{M}_{\mathrm{out}} = \mathcal{M}_{\mathrm{cont}} \cup \mathcal{M}_{\mathrm{term}}$ (fully formalized in Definition S3.4). Active periods satisfy $t < \tau$ while captivity continues.

DEFINITION S3.1—MDG stochastic implementation filter: Let $i \in \{K, S, F\}$ and let $\mathcal{A}^i$ be the finite set of admissible actions of agent i. The **state vector** is $X_t := (\mu_t, h_t)$, with μ_t from

Definition S2.3 and h_t the public history at the opening of t. Given the plan $a_t^{i*} \in \mathcal{A}^i$ and X_t, execution is determined by the **implementation law**

$$\tilde{a}_t^i \sim \mathbb{P}_{\mathrm{I},i}\big(\cdot \mid a_t^{i*},\, X_t\big).$$

In the base logit specification (S3.3), $\mathbb{P}_{\mathrm{I},i}(\cdot \mid a_t^{i*}, X_t)$ depends on X_t only through μ_t (via the temperature T_t); conditioning on h_t is reserved for extended specifications. The stochastic structure is formalized on a measurable space $(\Omega, \mathcal{F})$. The measurable function $\mathcal{H}_t^i : \mathcal{A}^i \times \Omega \to \mathcal{A}^i$ materializes the sequential draw: $\tilde{a}_t^i(\omega) = \mathcal{H}_t^i(a_t^{i*}, \omega)$. The **aggregate profile of executed actions** is

$$\tilde{A}_t = (\tilde{a}_t^K,\, \tilde{a}_t^S,\, \tilde{a}_t^F). \tag{S3.2}$$

The following regularity assumptions are imposed.

(i) **Plan dominance and error support.** In the **base specification** with $\underline{c} > 0$, for all $t < \tau$ (with τ as in (S3.1)), $\mathbb{P}_{\mathrm{I},i}$ satisfies strict **plan dominance**, $\mathbb{P}_{\mathrm{I},i}(a_t^{i*} \mid a_t^{i*}, X_t) > \mathbb{P}_{\mathrm{I},i}(a \mid a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$, and **full support**, $\mathbb{P}_{\mathrm{I},i}(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.

(ii) **Bounded institutional maturation.** The informational component $[H(\mu_t)/H(\mu_0)]e^{-\eta_{\mathrm{cal}} t}$ declines as posterior uncertainty falls, while the structural floor $\underline{c}$ persists in (S3.4). When $\underline{c} > 0$, even if $H(\mu_t) \to 0$ one has $T_t \geq T_0 \underline{c}$ and $\mathbb{P}_{\mathrm{I},i}(\tilde{a}_t^i = a_t^{i*} \mid a_t^{i*}, X_t) < 1$, so full support over $\mathcal{A}^i$ is preserved in the long run.

The implementation law adopts the **hybrid-temperature logit** structure:

$$\mathbb{P}_{\mathrm{I},i}\big(\tilde{a}_t^i = a \mid a_t^{i*}, X_t\big) = \frac{\exp\big(\mathbf{1}\{a = a_t^{i*}\}/T_t\big)}{\displaystyle\sum_{a' \in \mathcal{A}^i} \exp\big(\mathbf{1}\{a' = a_t^{i*}\}/T_t\big)}, \tag{S3.3}$$

with hybrid temperature

$$T_t = T_0 \max\Big\{ \frac{H(\mu_t)}{H(\mu_0)} e^{-\eta_{\mathrm{cal}} t},\, \underline{c} \Big\}, \tag{S3.4}$$

where $T_0 > 0$ is the initial temperature, $\eta_{\mathrm{cal}} > 0$ is the calendar decay rate, Shannon entropy is $H(\mu_t) = -\sum_\theta \mu_t(\theta) \ln \mu_t(\theta)$ with $H(\mu_0) > 0$ by the full-support prior in Definition S2.3, and $\underline{c} \in [0,1]$ is the floor of structural noise.

□

REMARK S3.2—Residual noise and the limit $\underline{c} = 0$: The special case $\underline{c} = 0$ is interpreted as a limit without residual noise: it allows $\mathbb{P}_{\mathrm{E}}(\tilde{a}_t^i = a_t^{i*} \mid \mathcal{F}_t^{\mathrm{pub}}) \to 1$ when $H(\mu_t) \to 0$, but sacrifices the persistent full support maintained under $\underline{c} > 0$. The base version with $\underline{c} > 0$ aligns with the absolute continuity or "grain of truth" condition of **?**, according to which rational learning must not assign zero probability to events that can occur.

REMARK S3.3—Axiomatic justification of the logit as the unique implementation law: Equation (S3.3) is the *unique* distribution over $\mathcal{A}^i$ that simultaneously satisfies the three axioms:

(A1) **Full support:** $\mathbb{P}_{\mathrm{I},i}(a \mid a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.

(A2) **Dominance:** $\mathbb{P}_{\mathrm{I},i}(a_t^{i*} \mid a_t^{i*}, X_t) > \mathbb{P}_{\mathrm{I},i}(a \mid a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$.

(A3) **Maximum entropy:** Among all distributions satisfying (A1) and (A2), $\mathbb{P}_{\mathrm{I},i}$ maximizes the Shannon entropy $H = -\sum_a p_a \ln p_a$ subject to the optimal action receiving fixed probability $1 - \varepsilon_t \in (0,1)$, where $\varepsilon_t := 1 - \mathbb{P}_{\mathrm{I},i}(a_t^{i*} \mid a_t^{i*}, X_t)$ is the implementation-error rate induced by the logit.

In the parametrization (S3.3), ε_t is decreasing in $1/T_t$. By the characterization theorem of **?**, the unique maximizer is the logit in (S3.3). The MDG process is thus axiomatically justified as the unique implementation process compatible with bounded rationality and maximum entropy.

DEFINITION S3.4—Law of the physical outcome: The **physical outcome** m_t is an observable categorical variable on the support $\mathcal{M}_{\text{out}}$, where $\mathcal{M}_{\text{cont}} = \{\text{cont}\}$, $\mathcal{M}_{\text{term}} = \{\text{pay}, \text{res}, \text{rel}, \text{kill}\}$ and $\mathcal{M}_{\text{out}} = \mathcal{M}_{\text{cont}} \cup \mathcal{M}_{\text{term}}$. The label cont represents the continuation of captivity and $j \in \{1,2,3,4\}$ closure by competitive cause j. For $j \in \{1,2,3\}$, the effective intensity $\tilde{\lambda}_j(t \mid \theta_K)$ carries the type-specific parameter $\beta_{K,j}(\theta_K)$ in the Cox specification; cause $j = 4$ enters only through the baseline hazard (there is no $\beta_{K,4}$). Conditional on the captor's type and the material state of the period

$$\mathcal{C}_t(\theta_K) = \big(\theta_K, \theta_F, \theta_V, \theta_S, z, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, \alpha_t^*, \gamma_t^*, p_{\det,t}\big),$$

the distribution of m_t is defined from the effective intensities $\tilde{\lambda}_j(t \mid \theta_K)$. Denote by

$$Q_t(m \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*) := \mathbb{P}_{\mathrm{E}}(m_t = m \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*)$$

the transition kernel of the physical outcome, and by G_t the measurable rule that materializes a concrete realization of m_t from that kernel and a uniform draw. The **continuation probability** is

$$Q_t(\text{cont} \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*) = \exp\left(-\sum_{j=1}^{4} \tilde{\lambda}_j(t \mid \theta_K)\,\Delta t\right).$$

The daily closure total and the relative weight of each cause are

$$q_t(\theta_K) = 1 - p_{\text{Cont},t}, \quad \xi_j = \frac{\tilde{\lambda}_j}{\sum_{\ell} \tilde{\lambda}_\ell}, \quad \bar{h}_j(t \mid \theta_K) = q_t\,\xi_j.$$

The materialization of m_t is obtained by **inverse transform**: let $v_t \sim \text{Unif}(0,1)$; then

$$m_t = r \iff \sum_{\ell < r} Q_t(\ell \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*) \le v_t < \sum_{\ell \le r} Q_t(\ell \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*),$$

and the likelihood of m_t in the Bayesian update is

$$\mathbb{P}_{\mathrm{E}}(m_t \mid \theta_K, \mathcal{C}_t) = p_{\text{Cont}}^{\mathbf{1}\{m_t = \text{cont}\}} \prod_{j=1}^{4} \bar{h}_j^{\mathbf{1}\{m_t = j\}}.$$

□

S4. ASYMPTOTIC TYPE IDENTIFICATION

This section defines the prolonged captivity event and the notion of asymptotic identification, states the structural identifiability lemma (bridge between the parameters $\beta_{K,j}$ and the singularity of laws), and concludes with the posterior consistency theorem.

Notation. Throughout Sections S4 and S8, θ_K^* denotes the realized captor type. When conditioning on the law of the true type, the notation $\theta^* := \theta_K^*$ and $\mu_C := \mathbb{P}_{\mathrm{C},\theta^*}(\cdot \mid \mathcal{T})$ is adopted.

DEFINITION S4.1—Prolonged captivity event: Let τ be as in (S3.1). The **prolonged captivity event** is

$$\mathcal{T} := \{\tau = \infty\} = \big\{\omega \in \Omega : m_t(\omega) \in \mathcal{M}_{\mathrm{cont}}\ \forall t \geq 0\big\}.$$

In $\mathcal{T}$ the episode admits no terminal outcome in any finite period and generates an infinite sequence of public signals $\{\zeta_t\}_{t\geq 0}$.

□

DEFINITION S4.2—Asymptotic identification in the event $\mathcal{T}$: The model is **asymptotically identifiable** in $\mathcal{T}$ (Definition S4.1) if $\mathbb{P}_{\mathrm{C},\theta_K^*}(\mathcal{T}) > 0$ and, for every $\theta \in \Theta_K$ with $\theta \neq \theta_K^*$, at least one of the following conditions holds.

(a) **Exclusion by survival.** $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) = 0$.

(b) **Tail singularity of signals.** $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) > 0$ and the laws of $\{\zeta_t\}_{t\geq 0}$ restricted to $\mathcal{T}$ under $\mathbb{P}_{\mathrm{C},\theta}$ and under $\mathbb{P}_{\mathrm{C},\theta_K^*}$ are **mutually singular**.

□

REMARK S4.3—Role of Lemma S4.4 in the analysis: Theorem S4.6 requires asymptotic identification (Definition S4.2), a condition formulated in terms of laws of entire processes. Lemma S4.4 is the **operational bridge** between that abstract hypothesis and the parameters $\beta_{K,j}$ already introduced in the mechanism: it allows one to verify in the practice of the model when two distinct types generate asymptotically distinguishable trajectories. Without this result, the learning section would rest on a high-level condition difficult to connect with the proportional parametrization of the λ_j.

LEMMA S4.4—Structural identifiability and asymptotic learning: *Suppose $|\Theta_K| < \infty$. Let $Y_t := (m_t, d_t)$ be the daily public record. Under a PBE, $p_\theta(\cdot \mid \mathcal{F}_t^{\mathrm{pub}})$ denotes the mass function of Y_t conditioned on public information at the opening of t under type θ.*

Part (i) — local sufficiency (??). *Let $\theta, \theta' \in \Theta_K$ be distinct. If there exists $j \in \{1,2,3\}$ such that $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ (causes $j \in \{1,2,3\}$ only; see Definition S3.4) and no exact compensations arise between the masses of Y_t (*knife-edges*), then there exists $\varepsilon = \varepsilon(\theta, \theta', t) > 0$ such that*

$$\big\|p_\theta(\cdot \mid \mathcal{F}_t^{\mathrm{pub}}) - p_{\theta'}(\cdot \mid \mathcal{F}_t^{\mathrm{pub}})\big\|_{\mathrm{TV}} \geq \varepsilon. \tag{S4.1}$$

Part (ii) — asymptotic sufficiency (??). *Fixing a policy trajectory $(\chi_s)_{s\geq 0}$, suppose that conditional on $\mathcal{T}$ the $\{Y_t\}_{t\geq 0}$ are independent. For each pair $\theta \neq \theta'$, let $\mathcal{S}_{\theta,\theta'} \subseteq \mathbb{N}$ be the set of periods where (S4.1) holds. If $\mathcal{S}_{\theta,\theta_K^*}$ is infinite $\mathbb{P}_{\mathrm{C},\theta_K^*}$-a.s. and there exists $\underline{\varepsilon}_\theta > 0$ such that $\|p_\theta^{(t)} - p_{\theta_K^*}^{(t)}\|_{\mathrm{TV}} \geq \underline{\varepsilon}_\theta$ for all $t \in \mathcal{S}_{\theta,\theta_K^*}$, then the laws of $(Y_t)_{t\geq 0}$ conditioned on $\mathcal{T}$ under $\mathbb{P}_{\mathrm{C},\theta}$ and under $\mathbb{P}_{\mathrm{C},\theta_K^*}$ are **mutually singular**.*

REMARK S4.5—Informational Slicing and Conditional Independence: A crucial conceptual distinction must be maintained between the adaptive equilibrium path of the active dual control policy and the measure-theoretic construction required by Lemma S4.4 (**??**). In the live perfect Bayesian equilibrium, the planner updates beliefs sequentially, causing the realized instruments (α_t^*, γ_t^*) to be non-trivially coupled with the historical filtration $\{\mathcal{F}_t^{\text{pub}}\}$ (**??**). However, the probability measures $\mathbb{P}_{C,\theta}$ invoked in Definition S4.2 and Lemma S4.4 operate on a frozen deterministic policy path $(\chi_s)_{s\geq 0}$ evaluated ex-ante at $t = 0$ (**?**). Within this sliced counterfactual regime, the strategic intentions of the private agents are uniquely pinned down by sequential rationality, rendering the daily innovations of the physical-institutional block $Y_t = (m_t, d_t)$ conditionally independent across periods (**?**). This conditional factorization satisfies the orthogonal product-measure structure $\left(\bigotimes_{t\geq 0} p_\theta^{(t)}\right)$ required to validly operationalize the singularity criterion of **?** without violating the temporal feedback loop of the dynamic game.

PROOF: The proof identifies the causal mechanism within the day, translates it to a statistical bound in total variation, and couples the repeated experiment to the criterion of **?**.

1. Same control menu, distinct criminal technology. Fix t and $\mathcal{F}_t^{\text{pub}}$. In the counterfactual that changes only the type (θ vs. θ'), keeping equal the executed triple $\tilde{A}_t$, the instruments (α_t^*, γ_t^*), and the maturation filter $M(t)$, the proportional parametrization implies

$$\frac{\lambda_j(t \mid \theta)}{\lambda_j(t \mid \theta')} = \exp\bigl(\beta_{K,j}(\theta) - \beta_{K,j}(\theta')\bigr) \neq 1.$$

2. From daily intensity to distinct predictions over Y_t. The proportional difference in $\lambda_j(t)$ is projected onto the aggregate exponential of $q(t)$, the causal mix $\xi_j(t)$, or the marginal hazards $\bar{h}_j(t) = q(t)\xi_j(t)$. Except for *knife-edges*, the marginal of m_t cannot coincide under θ and θ', and the two mass functions differ in total variation, verifying (S4.1).

3. Repetition with uniform margin and Kakutani criterion. Conditional on the fixed policy path and on $\mathcal{T}$—under the informational slicing regime clarified in Remark S4.5—the law of the sequence factors exactly as a product of daily independent experiments $\bigotimes_{t\geq 0} p_\theta^{(t)}$ versus $\bigotimes_{t\geq 0} p_{\theta_K^*}^{(t)}$ (**?**). Part (ii) then follows directly from Lemma S8.1 (Section S8): recurrent uniform total-variation separation on an infinite set forces the Bhattacharyya product affinity to zero and, by the criterion of **?**, mutual singularity of the infinite product measures (**?**). *Q.E.D.*

THEOREM S4.6—Identification and consistency of true types: *Consider a prior $\mu_0 \in \Delta(\Theta_K)$ with full support and a PBE-consistent Bayesian updating (Definition S2.3). Let $\theta^* := \theta_K^*$ be the realized type and $\mu_C := \mathbb{P}_{C,\theta^*}(\cdot \mid \mathcal{T})$ the law of the true type restricted to the event $\mathcal{T}$ (Definition S4.1).*

*If the mechanism is **asymptotically identifiable** in $\mathcal{T}$ (Definition S4.2), the following properties are satisfied μ_C-almost surely:*

1. ***Exclusion of false hypotheses:*** $\mu_t(\theta) \xrightarrow{t\to\infty} 0$ *for all* $\theta \neq \theta^*$.
2. ***Concentration on the true type:*** $\mu_t(\theta^*) \xrightarrow{t\to\infty} 1$.

Consequently, in $\mathcal{T}$ the sequence $\{\mu_t\}$ converges weakly to the Dirac measure δ_{θ^*}.

PROOF: The proof separates the general probabilistic argument from the structural identification assumption.

S4.0.0.1. *1. Stability of beliefs.* By equilibrium consistency, $\mu_t(\theta) = \mathbb{E}^{\mathbb{P}_{\mathrm{E}}}[\mathbf{1}\{\theta_K = \theta\} \mid \mathcal{F}_t^{\mathrm{pub}}]$ $\mathbb{P}_{\mathrm{E}}$-a.s. Lemma S2.4 guarantees that $\{\mu_t(\theta)\}$ is a bounded martingale with limit $Z_\theta \in [0,1]$ $\mathbb{P}_{\mathrm{E}}$-a.s. The transfer to μ_C-a.s. is justified by disintegration: the mixture $\mathbb{P}_{\mathrm{E}} = \sum_\vartheta \mu_0(\vartheta)\mathbb{P}_{\mathrm{C},\vartheta}$ implies that every $\mathbb{P}_{\mathrm{E}}$-null set is $\mathbb{P}_{\mathrm{C},\theta^*}$-null (since $\mu_0(\theta^*) > 0$), and domination is preserved upon restricting to $\mathcal{T}$.

S4.0.0.2. *2. Identification and exclusion of false types.* Fix $\theta \neq \theta^*$.

Case $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) = 0$: the trajectory under μ_C belongs to an event that the false type cannot generate; its posterior likelihood is null in the conditioned regime.

Case $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) > 0$: the singularity of Definition S4.2(b) implies that there exists a tail event A_θ with $\mathbb{P}_{\mathrm{C},\theta^*}(A_\theta \mid \mathcal{T}) = 1$ and $\mathbb{P}_{\mathrm{C},\theta}(A_\theta \mid \mathcal{T}) = 0$. Let R_t^θ be the likelihood ratio of the false type against the true type on finite horizons. Tail singularity forces $R_t^\theta \to 0$ μ_C-a.s. (**??**). By Bayes' formula with a full-support prior,

$$\mu_t(\theta) = \frac{\mu_0(\theta)\, R_t^\theta}{\mu_0(\theta^*) + \sum_{\theta' \neq \theta^*} \mu_0(\theta') R_t^{\theta'}} \leq \frac{\mu_0(\theta)}{\mu_0(\theta^*)}\, R_t^\theta,$$

and since $\mu_0(\theta^*) > 0$, it follows that $\mu_t(\theta) \to 0$ μ_C-a.s. for all $\theta \neq \theta^*$.

S4.0.0.3. *3. Closure by normalization.* For all t, $\sum_{\theta \in \Theta_K} \mu_t(\theta) = 1$. As Θ_K is finite and all false coordinates converge to zero μ_C-a.s.,

$$1 = \lim_{t\to\infty} \sum_\theta \mu_t(\theta) = Z_{\theta^*} + \sum_{\theta \neq \theta^*} 0,$$

hence $Z_{\theta^*} = 1$ μ_C-a.s., that is, $\mu_t(\theta^*) \to 1$ μ_C-a.s. On a finite simplex, coordinate convergence is equivalent to weak convergence $\mu_t \to \delta_{\theta^*}$. *Q.E.D.*

REMARK S4.7—Economic reading and link with $\beta_{K,j}$: Definition S4.2 requires that the design of the game (policy + incentives + risk technology) produce long trajectories incompatible with false types. Distinct shifts $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ favor the singularity of the laws of (m_t, d_t). If the equilibrium pools types, Lemma S2.4 still holds but Theorem S4.6 does not: the limit may split mass among observationally equivalent types.

REMARK S4.8—Finite horizon: If $\tau < \infty$ almost surely, $\mathcal{T}$ has measure zero and the statement of Theorem S4.6 is vacuous. The relevant conclusion is then only the martingale convergence of Lemma S2.4: beliefs stabilize at the finite limit of observations.

S5. COMPETITIVE RISKS AND OPERATIONAL PRESSURE

The following proposition uses directly the proportional risk structure, without requiring the equilibrium machinery of Section S7. It depends only on the intensities $\tilde{\lambda}_j$ introduced in Definition S3.4.

PROPOSITION S5.1—Expected duration of captivity and operational pressure: *Fix a type θ_K and a period t. The following standing assumptions govern the comparative static in γ_t^*.*

Standing assumptions. (S0) *Baseline hazards satisfy $\lambda_{j0}(s) > 0$ for all strategic causes $j \in \{1,2,3\}$ and all s in the support, so $\mathbb{E}[\tau \mid \theta_K] < \infty$.* (S1) *The object $\partial\mathbb{E}[\tau \mid \theta_K]/\partial\gamma_t^*$ is a partial* *derivative: only γ_t^* is perturbed at period t while $\{\gamma_s^*\}_{s\neq t}$ is held fixed. Writing $S(s \mid \theta_K) := \prod_{r=1}^{s} p_{\mathrm{Cont},r|\theta_K}$ with $S(0 \mid \theta_K) = 1$, expected duration admits the survival representation*

$$\mathbb{E}[\tau \mid \theta_K] = \sum_{s=0}^{\infty} S(s \mid \theta_K), \tag{S5.1}$$

so only factors with $s \geq t$ depend on γ_t^*. (S2) *For each strategic cause $j \in \{1,2,3\}$,*

$$\frac{\partial \tilde{\lambda}_j(t \mid \theta_K)}{\partial \gamma_t^*} = \eta_{\gamma,j}\, \tilde{\lambda}_j(t \mid \theta_K), \qquad \eta_{\gamma,j} := \frac{\partial \ln \tilde{\lambda}_j(t \mid \theta_K)}{\partial \gamma_t^*};$$

the exogenous cause $j = 4$ is invariant to γ_t^*.

Net sensitivity. In the main-paper hazard index, payment ($j = 1$) is decreasing in γ and death/state rescue ($j \in \{2,3\}$) are increasing, so $\eta_{\gamma,1} < 0$ and $\eta_{\gamma,j} > 0$ for $j \in \{2,3\}$. Let $\zeta_{\gamma,j} := |\eta_{\gamma,j}|$ and define

$$\kappa_h(\theta_K,t) := \zeta_{\gamma,2}\, \tilde{\lambda}_2(t \mid \theta_K) + \zeta_{\gamma,3}\, \tilde{\lambda}_3(t \mid \theta_K) - \zeta_{\gamma,1}\, \tilde{\lambda}_1(t \mid \theta_K) = \sum_{j=1}^{3} \eta_{\gamma,j}\, \tilde{\lambda}_j(t \mid \theta_K). \tag{S5.2}$$

Result. Under (S0)–(S2)*,*

$$\frac{\partial\, \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*} = -\kappa_h(\theta_K,t)\, \Delta t \sum_{s=t}^{\infty} S(s \mid \theta_K). \tag{S5.3}$$

In particular, whenever $\kappa_h(\theta_K,t) \neq 0$,

$$\operatorname{sgn}\!\left(\frac{\partial\, \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*}\right) = -\operatorname{sgn}(\kappa_h(\theta_K,t))\,. \tag{S5.4}$$

If $\kappa_h(\theta_K,t) > 0$, the one-shot perturbation in (S1) *lowers expected duration; if $\kappa_h(\theta_K,t) < 0$, it raises it. Cross-type variation in the sign enters through $\beta_{K,j}(\theta_K)$ and the induced semi-elasticities $\{\eta_{\gamma,j}\}$.*

PROOF: The argument has three steps: differentiate the daily continuation probability, propagate the perturbation through the survival product, and differentiate the duration identity (S5.1).

S5.0.0.1. *Step 0. Survival representation.* For a discrete-time competing-risks process with daily continuation probability $p_{\mathrm{Cont},s|\theta_K}$, the survival function $S(t' \mid \theta_K) = \prod_{s=1}^{t'} p_{\mathrm{Cont},s|\theta_K}$ satisfies $S(t' \mid \theta_K) = \mathbb{P}(\tau > t' \mid \theta_K)$. Hence

$$\mathbb{E}[\tau \mid \theta_K] = \sum_{t'=0}^{\infty} \mathbb{P}(\tau > t' \mid \theta_K) = \sum_{t'=0}^{\infty} S(t' \mid \theta_K),$$

which is the discrete-time survival identity (Wald's representation; see, e.g., **?**). Finiteness in (S0) makes the sum well defined.

S5.0.0.2. *Step 1. Daily continuation and κ_h.* By Definition S3.4,

$$p_{\mathrm{Cont},s|\theta_K} = \exp\Big(-\sum_{j=1}^{4} \tilde{\lambda}_j(s \mid \theta_K)\,\Delta t\Big) = \exp\big(-[\Lambda_s(\theta_K) + \tilde{\lambda}_4(s)]\,\Delta t\big),$$

where $\Lambda_s(\theta_K) := \sum_{j=1}^{3} \tilde{\lambda}_j(s \mid \theta_K)$ is the strategic hazard total. Only Λ_t depends on γ_t^*; the exogenous term $\tilde{\lambda}_4(t)$ is γ-invariant. By (S2),

$$\frac{\partial \Lambda_t(\theta_K)}{\partial \gamma_t^*} = \sum_{j=1}^{3} \eta_{\gamma,j}\,\tilde{\lambda}_j(t \mid \theta_K) = \kappa_h(\theta_K, t),$$

using $\eta_{\gamma,1} = -\zeta_{\gamma,1}$ and $\eta_{\gamma,j} = +\zeta_{\gamma,j}$ for $j \in \{2,3\}$ in (S5.2). Since $p_{\mathrm{Cont},t|\theta_K} = \exp(-\Lambda_t \Delta t) \cdot \exp(-\tilde{\lambda}_4(t)\Delta t)$ and the second factor is constant in γ_t^*,

$$\frac{\partial p_{\mathrm{Cont},t|\theta_K}}{\partial \gamma_t^*} = \exp(-\tilde{\lambda}_4(t)\Delta t) \cdot \frac{\partial}{\partial \gamma_t^*} \exp(-\Lambda_t \Delta t) = -\kappa_h(\theta_K, t)\, p_{\mathrm{Cont},t|\theta_K}\,\Delta t. \tag{S5.5}$$

S5.0.0.3. *Step 2. Propagation to accumulated survival.* Fix $t' \geq t$. Because only the factor $s = t$ in $S(t' \mid \theta_K) = \prod_{s=1}^{t'} p_{\mathrm{Cont},s|\theta_K}$ depends on γ_t^*,

$$\frac{\partial S(t' \mid \theta_K)}{\partial \gamma_t^*} = \Big(\prod_{\substack{s=1 \\ s \neq t}}^{t'} p_{\mathrm{Cont},s|\theta_K}\Big) \frac{\partial p_{\mathrm{Cont},t|\theta_K}}{\partial \gamma_t^*} = -\kappa_h(\theta_K, t)\, S(t' \mid \theta_K)\,\Delta t.$$

For $t' < t$, the derivative is zero.

S5.0.0.4. *Step 3. Expected duration and sign.* Differentiating (S5.1) term by term under (S1) and applying Step 2 yields (S5.3). The interchange of sum and derivative is justified by dominated convergence: for $t' \geq t$, $|\partial S(t' \mid \theta_K)/\partial \gamma_t^*| \leq |\kappa_h(\theta_K, t)|\, S(t' \mid \theta_K)\,\Delta t$, and

$$\sum_{t'=t}^{\infty} S(t' \mid \theta_K) \leq \sum_{t'=0}^{\infty} S(t' \mid \theta_K) = \mathbb{E}[\tau \mid \theta_K] < \infty$$

by (S0). Because $S(t' \mid \theta_K) \geq 0$ and the tail $\sum_{t'=t}^{\infty} S(t' \mid \theta_K) > 0$ whenever continuation has positive mass on $t' \geq t$, the sign of (S5.3) is the opposite of the sign of $\kappa_h(\theta_K, t)$. *Q.E.D.*

REMARK S5.2—Local comparative static and testable content: Proposition S5.1 is a *local* comparative static: it evaluates a one-period perturbation of γ_t^* holding the remaining policy path fixed. It does not claim global monotonicity of $\mathbb{E}[\tau \mid \theta_K]$ as a function of the entire sequence $\{\gamma_s^*\}_{s \geq 0}$. The empirical content is the sign prediction: types with larger κ_h respond differently to operational pressure because the Cox coefficients $\beta_{K,j}(\theta_K)$ shift the semi-elasticities $\{\eta_{\gamma,j}\}$ and the levels $\{\tilde{\lambda}_j\}$.

S6. POLICY IMPLICATIONS

This proposition requires the belief machinery of Section S2 and the equilibrium context of Section S7: optimal pressure is calibrated in a Bayesian manner.

PROPOSITION S6.1—Optimal operational pressure and Bayesian type identification: *Suppose $\Theta_K = \{\theta_{low}, \theta_{high}\}$ and fix a discrete branch $b \in \{R, N\}$ in a neighborhood of the posterior belief under consideration. Let $x_t = (\alpha_t, \gamma_t)$ and define the complete branch objective*

$$J_t^b(x_t, \mu_t) := \sum_{\theta \in \Theta_K} \mu_t(\theta) L_t(b, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,b}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t). \tag{S6.1}$$

If (α_t^*, γ_t^*) is an interior optimum of $J_t^b(\cdot, \mu_t)$, the constraints IC^K, IR^K, and IR^F do not locally change regime, the Hessian $H_t^b := \nabla_{xx}^2 J_t^b(x_t^*, \mu_t)$ is positive definite, and the net marginal effect of increasing $\mu_t(\theta_{high})$ on the first-order condition for γ_t favors greater pressure, that is,

$$e_\gamma' \left(H_t^b\right)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu_t) < 0, \qquad e_\gamma = (0,1)', \tag{S6.2}$$

then the optimal operational pressure of that branch is locally increasing in the posterior belief on the high-risk type:

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} > 0.$$

The structural separation $\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})$ for $j \in \{2,3\}$ contributes to this condition through the ratio

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp\!\big(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})\big), \quad j \in \{2,3\}.$$

Monotonicity, however, depends on the joint net effect of material losses, deviation penalties, and informational gains.

PROOF: Write $\mu := \mu_t(\theta_{high})$ and $\mu_t(\theta_{low}) = 1 - \mu$. Within the fixed discrete branch b, the interior optimum $x_t^*(\mu) = (\alpha_t^*(\mu), \gamma_t^*(\mu))$ satisfies

$$\nabla_x J_t^b(x_t^*(\mu), \mu) = 0. \tag{S6.3}$$

The hypothesis $H_t^b = \nabla_{xx}^2 J_t^b(x_t^*, \mu) \succ 0$ implies that the local optimum is isolated and that the Jacobian of the first-order condition with respect to x_t is invertible. By the implicit function theorem, $x_t^*(\mu)$ is locally differentiable and

$$\frac{dx_t^*}{d\mu} = -\left(H_t^b\right)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu). \tag{S6.4}$$

Taking the second coordinate yields

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} = -e_\gamma' \left(H_t^b\right)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu_t). \tag{S6.5}$$

By the net monotonicity condition (S6.2), the right-hand side of (S6.5) is strictly positive.

To interpret this condition, note that the coordinate of $\nabla_{x\mu}^2 J_t^b$ associated with γ_t contains

$$\frac{\partial L_t}{\partial \gamma_t}(b, \alpha_t^*, \gamma_t^*, \theta_{high}, \iota_t) - \frac{\partial L_t}{\partial \gamma_t}(b, \alpha_t^*, \gamma_t^*, \theta_{low}, \iota_t) + \frac{\partial}{\partial \mu}\left[\frac{\partial \Pi_{t,b}^S}{\partial \gamma_t} - \psi_H \frac{\partial \Delta H_t}{\partial \gamma_t}\right]_{x_t = x_t^*}. \tag{S6.6}$$

The first differential captures the material separation between types. Under proportional risks,

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp\bigl(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})\bigr),$$

so that greater structural distance in $j \in \{2,3\}$ increases the difference in marginal losses associated with operational pressure. However, the deviation premium and the informational subsidy also enter the first-order condition; therefore the result is formulated as monotonicity of the net marginal effect of the complete objective, not as a property of material loss alone. Under (S6.2), increasing the posterior mass on the high-risk type shifts the interior optimum toward a higher γ_t^*. *Q.E.D.*

S7. EQUILIBRIUM AND IMPLEMENTABILITY

This section introduces the dynamic mechanism with its PBE, the notion of implementability with the constraints IC^K, IR^K, IR^F, and proves the existence of the implementable equilibrium.

DEFINITION S7.1—Dynamic mechanism and Perfect Bayesian Equilibrium: Consider a discrete-time Bayesian game with horizon $\mathbb{T} = \{0, 1, \dots, \tau\}$, players $\mathcal{J} = \{S, K, F\}$, and captor type $\theta_K \in \Theta_K$ according to common prior μ_0 with $\mu_0(\theta) > 0$ for all θ. A **dynamic mechanism** is described by:

- *Strategy profile:* $s = (s^S, s^K, s^F)$, where $s^S = \chi = (\chi_t)_{t \in \mathbb{T}}$ with $\chi_t : H_t \times \mathcal{R} \to \mathcal{A}^{S,\mathrm{disc}} \times \mathcal{A}^{\alpha} \times \mathcal{A}^{\gamma}$.
- *Belief system:* $\mu = \{\mu_t\}_{t \geq 0}$, $\mu_t \in \Delta(\Theta_K)$.
- *Ex-ante law:* $\mathbb{P}^s_{\mathrm{E}}$ induced by profile s, prior μ_0, and the physical dynamics of captivity.

Perfect Bayesian Equilibrium. The pair (s, μ) is a **PBE** if the following hold:

(a) **Sequential rationality.** At every node reached with positive probability, each agent i solves its optimization program given μ_t and s^{-i}. For the state:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t)} \Bigl\{ \sum_{\theta} \mu_t(\theta) L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi^S_{t, a^S}(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Bigr\}.$$

For $i \in \{K, F\}$: $a_t^{i*} \in \arg\max_{a_t^i} \mathcal{U}_t^i(a_t^i \mid \mathcal{I}_t^i, \chi, s^{-i})$.

(b) **On-path Bayesian consistency.**

$$\mu_t(\theta) = \mathbb{P}^s_{\mathrm{E}}(\theta_K = \theta \mid \mathcal{F}_t^{\mathrm{pub}}), \qquad \forall \theta,\ t \geq 1.$$

□

REMARK S7.2: Definition S7.1 defines the PBE as an *object*: the pair (s, μ) satisfying (a) and (b). It does not assert existence nor that the profile satisfies incentive and participation constraints; those properties are the content of Theorem S7.6.

DEFINITION S7.3—Nash-Bayes-IC^K,IR^K,IR^F implementability: Let (s^*, μ^*) be a PBE (Definition S7.1). The mechanism is **implementable Nash-Bayes-IC^K,IR^K,IR^F** if the profile additionally satisfies the following conditions:

Bayesian incentive compatibility of the captor (IC^K). For all $t \in \mathbb{T}$ and all rival mimicry $\theta_j \in \Theta_K$,

$$\mathbb{E}_{\theta \sim \mu_t^*}\left[V_t^K\big(a^*(\theta) \mid \theta, \alpha_t^*, \gamma_t^*\big) - V_t^K\big(a^*(\theta_j) \mid \theta, \alpha_t^*, \gamma_t^*\big)\right] \geq 0.$$

Participation constraint of the captor (IR^K).

$$\mathbb{E}_{\theta \sim \mu_t^*}\left[U_{\text{rel}}^K(\theta) - \max\big\{V_{\text{cont},t}^K(\theta, \alpha_t^*, \gamma_t^*), U_{\text{kill}}^K(\theta, \theta_S)\big\}\right] \geq 0.$$

Participation constraint of the family (IR^F).

$$\mathcal{U}_t^F(a_{\text{coop}}) \geq \mathcal{U}_t^F(a_{\text{col}}).$$

The **feasible set of policies** is

$$\Gamma_t(\mu_t) := \left\{(a_t^S, \alpha_t, \gamma_t) \in \mathcal{A}^{S,\text{disc}} \times [0,1]^2 : IC^K,\ IR^K,\ IR^F \text{ are satisfied}\right\}. \tag{S7.1}$$

Its non-emptiness — **robust feasibility** — is the primitive hypothesis of Theorem S7.6.

□

REMARK S7.4: Definition S7.3 states implementability as a *property* of the pair (s^*, μ^*): given the existence of the PBE, one asks whether it additionally satisfies IC^K, IR^K, and IR^F. The cycle criterion of **?** provides a useful algebraic reformulation for auditing IC^K when Θ_K is finite.

REMARK S7.5—Full support of the MDG and Bayesian operability: The full support of $\mathbb{P}_{\text{I},i}$ (hypothesis (iii) of Theorem S7.6) guarantees strictly positive probability at all reachable nodes, making Bayesian updating operative on the path and reducing indeterminacy off the path.

THEOREM S7.6—Conditional implementability of the mechanism in PBE: *Consider the mechanism of Definition S7.1 under the following conditions:*

(i) $|\Theta_K| < \infty$ *with strictly positive prior.*
(ii) *Finite action spaces* $\mathcal{A}^i$ *for all* $i \in \{K, S, F\}$.
(iii) *The law* $\mathbb{P}_{\text{I},i}$ *of the MDG process assigns strictly positive probability to every action* ***(full support)***.
(iv) $\mathcal{A}^\alpha = \mathcal{A}^\gamma = [0,1]$, *compact and convex;* L_t *and* $\mathcal{U}_t^i$ *are measurable, bounded, and continuous in the continuous decision variables. The physical outcome rule* G_t *is measurable, and the transition kernel* Q_t *induced by* G_t *and by the competitive risk probabilities is continuous in those variables.*
(v) ***Robust feasibility.*** *For each* t *and* $\mu_t \in \Delta(\Theta_K)$, $\Gamma_t(\mu_t)$ *is non-empty and compact; the sections* $\Gamma_t^{a^S}(\mu_t)$ *are convex; the correspondence* $\mu_t \mapsto \Gamma_t(\mu_t)$ *is continuous with compact values.*

(vi) ***Parametric separability.***

$$V_t^K(a^*(\theta') \mid \theta_K, \alpha_t, \gamma_t) = \sum_j w_j(a^*(\theta'), \alpha_t, \gamma_t) \exp(\beta_{K,j}(\theta_K) \cdot z).$$

(vii) ***Quasiconvexity of the restricted objective.*** *For each t, μ_t, and a_t^S, the functional $(\alpha_t, \gamma_t) \mapsto \sum_\theta \mu_t(\theta) L_t + \Pi_{t,a^S}^S - \psi_H \Delta H_t$ is quasiconvex in (α_t, γ_t) over $\Gamma_t^{a^S}(\mu_t)$.*

Then there exists (s^*, μ^*) such that:

(a) *Constitutes a* ***PBE*** *(Definition S7.1).*

(b) ***Implements the social choice rule*** *f_t: $\chi_t^*(h_t) = (a_t^{S*}, \alpha_t^*, \gamma_t^*)$ minimizes the expected social loss subject to $\Gamma_t(\mu_t^*)$.*

(c) ***Satisfies*** *IC^K**,** IR^K**, and** IR^F in profile (s^*, μ^*).*

PROOF: The proof proceeds in five steps.

S7.0.0.1. *1. Existence of equilibrium.* By (ii), the behavioral strategy space Σ^i of each agent is a finite product of simplices, compact and convex. By (v), $\Gamma_t(\mu_t)$ is compact, non-empty, with convex sections and continuous correspondence. The state's objective is continuous in (α_t, γ_t) by (iv); by **Berge's maximum theorem** (**?**), the correspondence of optima $\Psi_t(\mu_t)$ is upper hemicontinuous and compact. By (vii), the functional is quasiconvex; the argmin of a quasiconvex function over a compact convex set is convex, so $\Psi_t(\mu_t)$ has convex values. The utilities of K and F are multilinear in behavioral strategies, so the best response correspondence

$$\mathrm{BR}(s) := \mathrm{BR}_\Gamma^S(s^{-S}) \times \mathrm{BR}^K(s^{-K}) \times \mathrm{BR}^F(s^{-F})$$

has non-empty convex values. By **Kakutani's fixed-point theorem** (**?**), there exists s^* with $s^* \in \mathrm{BR}(s^*)$.

S7.0.0.2. *2. On-path Bayesian consistency.* By (iii), $\mathbb{P}_{\mathrm{I},i}$ assigns positive probability to all actions compatible with the physical dynamics. Bayes' formula

$$\mu_t^*(\theta) = \mathbb{P}_{\mathrm{E}}^{s^*}\big(\theta_K = \theta \mid \sigma(z, \zeta_0, \ldots, \zeta_{t-1})\big)$$

is well defined at reached nodes and satisfies condition (b) of Definition S7.1.

S7.0.0.3. *3. Implementation of the social choice rule.* At the fixed point, the state's strategy solves at each node on the path:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t^*)} \Big\{ \sum_\theta \mu_t^*(\theta) L_t + \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t^*) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \Big\}.$$

The objective is continuous by (iv) and $\Gamma_t(\mu_t^*)$ is compact and non-empty by (v); by Weierstrass a solution exists.

S7.0.0.4. *4. Incentive compatibility (IC^K): cycle reformulation (**?**).* IC^K is imposed through the feasible set $\Gamma_t(\mu_t^*)$, not verified ex post: equilibrium instruments belong to $\Gamma_t(\mu_t^*)$, which by (v) and Definition S7.3 incorporates IC^K as a constitutive condition. For algebraic auditing with $\Delta_{ij}^t := V_t^K(a^*(\theta_j) \mid \theta_i) - V_t^K(a^*(\theta_i) \mid \theta_i)$, the Bayesian version is equivalent to $\mathbb{E}_{\theta_i \sim \mu_t^*}[\Delta_{ij}^t] \leq 0$ for all j. For every cycle $(\theta_{k_0}, \ldots, \theta_{k_m}, \theta_{k_0})$:

$$\sum_{\ell=0}^{m} \mu_t^*(\theta_{k_\ell})\, \Delta_{k_\ell, k_{\ell+1}}^t \leq 0 \qquad \text{(indices mod } m+1\text{)}.$$

S7.0.0.5. *5. Participation constraints (IR^K, IR^F).* By construction in Step 3, $(\alpha_t^*, \gamma_t^*) \in \Gamma_t(\mu_t^*)$, a set that by (v) simultaneously guarantees IR^K and IR^F in the equilibrium profile.

Having completed the five steps, (s^*, μ^*) is a PBE that implements f_t and satisfies IC^K, IR^K, and IR^F simultaneously. *Q.E.D.*

REMARK S7.7—Separation of results: implementability and asymptotic identification: Theorem S7.6 and Theorem S4.6 guarantee complementary properties. The first guarantees existence of a PBE with implementable policy subject to Γ_t: IC^K enters through the definition of the feasible set. The second guarantees that under that same equilibrium, if heterogeneity in $\beta_{K,j}$ generates uniform separation in total variation, beliefs converge to the true type almost surely in $\mathcal{T}$. The full support of the MDG plays a role in both: it reduces the *off-path* region and avoids degeneracies of the likelihood.

S8. ABSTRACT STRUCTURAL IDENTIFICATION (KAKUTANI CRITERION)

This section isolates the measure-theoretic argument: under product structure and recurrent separation in total variation, the criterion of **?** forces mutual singularity of trajectory laws. Theorem S8.2 states the same Bayesian conclusion as Section S4, replacing Lemma S4.4 with the abstract hypotheses of the following lemma.

LEMMA S8.1—Product of experiments and singularity (Kakutani criterion): *Fix $\theta \neq \theta'$. Suppose that conditional on equilibrium policy and on event $\mathcal{T}$, the sequence $\{Y_t\}_{t\geq 0}$ is independent and each Y_t takes values in a common finite support $\mathcal{Y}_t$. Let $\rho_t := \rho\big(p_\theta^{(t)}, p_{\theta'}^{(t)}\big)$ be the Bhattacharyya affinity. Then the affinity between the product laws on horizon T is*

$$\rho\Big(\bigotimes_{t=0}^{T} p_\theta^{(t)}, \bigotimes_{t=0}^{T} p_{\theta'}^{(t)}\Big) = \prod_{t=0}^{T} \rho_t. \tag{S8.1}$$

*If there exists an infinite set $\mathcal{S}_{\theta,\theta'} \subseteq \mathbb{N}$ with $\|p_\theta^{(t)} - p_{\theta'}^{(t)}\|_{\mathrm{TV}} \geq \underline{\varepsilon} > 0$ for all $t \in \mathcal{S}_{\theta,\theta'}$, then $\prod_{t\in\mathcal{S}} \rho_t = 0$ and, by the criterion of **?**, the product measures over the infinite sequence are* ***mutually singular*** *(**?**).*

PROOF: **Identity (S8.1).** Under independence,

$$\rho\Big(\bigotimes_{t=0}^{T} p_\theta^{(t)}, \bigotimes_{t=0}^{T} p_{\theta'}^{(t)}\Big) = \sum_{(y_0,\dots,y_T)} \prod_{t=0}^{T} \sqrt{p_\theta^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)} = \prod_{t=0}^{T} \sum_{y_t\in\mathcal{Y}_t} \sqrt{p_\theta^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)}.$$

Singularity. The proof uses the standard inequality $\|p-q\|_{\mathrm{TV}} \leq \sqrt{2}\,\mathsf{H}(p,q)$ on finite spaces, with $\mathsf{H}^2(p,q) = 1 - \rho(p,q)$. If $\|p-q\|_{\mathrm{TV}} \geq \underline{\varepsilon}$, then

$$\rho(p,q) \leq 1 - \frac{\underline{\varepsilon}^2}{2}.$$

At each $t \in \mathcal{S}_{\theta,\theta'}$, $\rho_t \leq 1 - \underline{\varepsilon}^2/2 < 1$. As $\mathcal{S}$ is infinite, the product contains infinitely many factors uniformly less than 1: $\prod_{t\in\mathcal{S}} \rho_t = 0$. By the criterion of **?**, the infinite product measures are mutually singular (**?**). *Q.E.D.*

THEOREM S8.2—Posterior consistency under product-measure singularity: *Suppose* Θ_K *finite, prior* μ_0 *with full support, and PBE-consistent Bayesian updating (Definition S2.3). Let* θ_K^* *be the realized type and* $\mathcal{T}$ *the prolonged captivity event (Definition S4.1). Suppose that for all* $\theta \neq \theta_K^*$ *the following hold:*

(i) ***Trajectory compatibility.*** $\mathbb{P}_{C,\theta_K^*}(\mathcal{T}) > 0$*; for each false type, either* $\mathbb{P}_{C,\theta}(\mathcal{T}) = 0$*, or signal laws are compared under the law restricted to* $\mathcal{T}$*.*

(ii) ***Product structure and recurrent TV separation.*** *The sequence* $\{Y_t\}_{t\geq 0}$ *satisfies product factorization and the existence of* $\mathcal{S}_{\theta,\theta_K^*}$ *and* $\underline{\varepsilon}_\theta > 0$ *from Lemma S8.1.*

(iii) ***Signal-experiment alignment.*** *In* $\mathcal{T}$*,* ζ_t *contains* $Y_t = (m_t, d_t)$*, so that singularity of* $\{Y_t\}$ *transfers to* $\{\zeta_t\}$ *by monotonicity of singularity with respect to* σ*-algebras.*

Then, in $\mathcal{T}$*,*

$$\mu_t(\theta) \xrightarrow{t\to\infty} 0 \ \forall\, \theta \neq \theta_K^*, \qquad \mu_t(\theta_K^*) \xrightarrow{t\to\infty} 1, \qquad \mathbb{P}_{C,\theta_K^*}\text{-a.s.}$$

Equivalently, $\{\mu_t\}$ *converges weakly to* $\delta_{\theta_K^*}$ $\mathbb{P}_{C,\theta_K^*}$*-a.s. in* $\mathcal{T}$*.*

PROOF: Fix $\theta \neq \theta_K^*$.

Case $\mathbb{P}_{C,\theta}(\mathcal{T}) = 0$: the false type cannot generate trajectories in $\mathcal{T}$; its posterior likelihood is annulled by survival exclusion.

Case $\mathbb{P}_{C,\theta}(\mathcal{T}) > 0$: by (ii), Lemma S8.1 implies that the infinite product laws are mutually singular. By (iii), that singularity transfers to the public flow $\{\zeta_t\}$. There exists a tail event A_θ with

$$\mathbb{P}_{C,\theta_K^*}(A_\theta \mid \mathcal{T}) = 1, \qquad \mathbb{P}_{C,\theta}(A_\theta \mid \mathcal{T}) = 0.$$

The likelihood ratio R_t^θ of the false type against the true type satisfies $R_t^\theta \to 0$ $\mathbb{P}_{C,\theta_K^*}$-a.s. in $\mathcal{T}$ (**??**). By Bayes,

$$\mu_t(\theta) \leq \frac{\mu_0(\theta)}{\mu_0(\theta_K^*)}\, R_t^\theta \to 0 \qquad \mathbb{P}_{C,\theta_K^*}\text{-a.s. in } \mathcal{T}.$$

As Θ_K is finite, all false coordinates converge to zero simultaneously. Finally,

$$1 = \lim_{t\to\infty} \sum_\theta \mu_t(\theta) = \lim_{t\to\infty} \mu_t(\theta_K^*) + 0,$$

so that $\mu_t(\theta_K^*) \to 1$ $\mathbb{P}_{C,\theta_K^*}$-a.s. in $\mathcal{T}$. *Q.E.D.*

S9. FORMAL VERIFICATION IN LEAN 4

This replication package includes a standalone Lean 4 project in folder `lean/` (main file `Appendix3Proofs.lean`) that formalizes the logical dependency graph of this supplement. The development is independent of earlier project-specific files and mirrors Sections S1–S8. Each declaration is tagged with its TeX label where applicable. A successful build (`lakebuild`) yields no `sorry` obligations. Build instructions appear in Section S10.

Reading the status tags. Every Lean declaration carries one of two tags. `[Proved]` means the statement is fully verified by the Lean 4 compiler against Mathlib: the proof is machine-checked and admits no gaps. `[Hypothesis]` means the statement is declared as an explicit interface and *assumed*, not re-proved, inside Lean; each such interface corresponds to a classical step proved analytically in this supplement or in the cited literature. The formalization therefore does *not* replace the proofs in this document: it certifies the finite-dimensional algebra and the logical assembly of the identification argument, and makes every classical input explicit.

Proved in Lean (machine-checked).

- **Section S1.** Lemma S1.1 and Corollary S1.2.
- **Section S2.** Public filtration, predictive likelihood, and martingale beliefs; Lemma S2.4 (bounded martingale with a.s. limit in $[0,1]$, via Doob's theorem in Mathlib); and the normalization closure of step 3 of Theorem S4.6: false-type collapse forces $\mu_t(\theta^*) \to 1$.
- **Section S3.** Continuation probability of Definition S3.4; Cox hazard-ratio algebra (distinct $\beta_{K,j} \Rightarrow$ ratio $\neq 1$); the predictive masses $p_\theta^{(t)}$ form probability masses; local TV separation — Lemma S4.4(i) — for those masses under $\beta_{K,j}$ separation and a non-knife-edge total hazard; logit axioms (A1) full support and (A2) plan dominance of Remark S3.3.
- **Section S5.** Proposition S5.1: compact and signed forms of κ_h (`kappa_h_signed`, `dLambda_eq_kappa`); continuation factorization and (S5.5) (`dContProb_eq`); sign certificate of (S5.3) with explicit Δt and survival tail (`prop_expected_captivity_gamma`, `prop_expected_captivity_sgn`). Wald identity and DCT remain analytic.
- **Section S4.** Logical assembly of Lemma S4.4: conditions (a)/(b) imply Definition S4.2; packaging of recurrent uniform TV margins; and the Bayes factorization of step 2 of Theorem S4.6: the bound $\mu_t(\theta) \leq (\mu_0(\theta)/\mu_0(\theta^*))\, R_t^\theta$ forces $\mu_t(\theta) \to 0$ whenever $R_t^\theta \to 0$.
- **Section S8.** Bhattacharyya product identity (S8.1); the Hellinger–TV bound $\rho \leq 1 - \underline{\varepsilon}^2/2$; and vanishing of the product affinity under recurrent separation, i.e., the entire algebraic part of Lemma S8.1.
- **Identification bridge.** Cox predictive masses plus a recurrent TV margin imply asymptotic identification, conditional only on the two Kakutani interfaces listed below.

Not proved in Lean (explicit hypotheses). The following classical steps are stated as interfaces and assumed inside Lean, mainly because infinite product measures and their convergence theory are not available in Mathlib. Each one is proved analytically where indicated.

- *Kakutani mutual singularity* — vanishing product affinity implies mutually singular trajectory laws: proof of Lemma S8.1 and **?**.
- *Singularity transfer* $Y_t \Rightarrow \zeta_t$: item (iii) of Theorem S8.2.
- *Singularity implies vanishing likelihood ratio*: step 2 of Theorem S4.6 (**??**).
- *Survival exclusion*, $\mathbb{P}_{\mathrm{C},\theta}(\mathcal{T}) = 0$ implies collapse: step 2 of Theorem S4.6.
- *Maximum entropy axiom (A3)* of Remark S3.3: **?**.
- *Implicit-function step* of Proposition S6.1: proof in Section S6.
- *Existence of the implementable PBE*, Theorem S7.6: fixed-point proof in Section S7 (**?**).

Table S9.1 maps these hypotheses to their Lean identifiers.

S10. REPRODUCIBILITY

All analytical proofs in this supplement are self-contained in `tex/Appendix_3.tex`. Machine-checked algebraic steps are reproduced by the Lean 4 project in `lean/`. Set the working directory to `Appendix_3/` and run `replicate.sh`. See `README.txt` at the package root for step-by-step instructions. Requirements: Lean 4 (toolchain in `lean-toolchain`) and Lake; Mathlib is fetched automatically on first build (network required; estimated first-build time 15–45 minutes).

1. `replicate.sh` at the package root runs `lakebuild` in `lean/` and writes a log to `lean/build.log`.
2. Main formal file: `lean/Appendix3Proofs.lean`. Project metadata: `lakefile.lean`, `lean-toolchain`, `lake-manifest.json`.

TABLE S9.1

LEAN FORMALIZATION STATUS FOR SELECTED SUPPLEMENT OBJECTS

TeX object	Lean identifier	Status
Lemma S8.1 (algebraic part)	`generalIdAppendixLemma`	Proved
Kakutani (1948) mutual singularity	`kakutaniMutualSingularity`	Hyp.
Outcome-trajectory laws on $\mathcal{T}$	`kakutaniSignalSingularity`	Hyp.
Singularity transfer $Y_t \Rightarrow \zeta_t$	`signalExperimentAlignment`	Hyp.
Singularity $\Rightarrow$ vanishing likelihood ratio	`singularityImpliesLikelihoodVanishing`	Hyp.
$\mathbb{P}_{C,\theta}(\mathcal{T}) = 0 \Rightarrow$ collapse	`zeroMassPosteriorCollapse`	Hyp.
Remark S3.3, axiom (A3)	`logitProb_maxEntropy_hyp`	Hyp.
Proposition S5.1, eq. (S5.2) (signed form)	`kappa_h_signed`, `dLambda_eq_kappa`	Proved
Proposition S5.1, eq. (S5.5)	`dContProb_eq`	Proved
Proposition S5.1, eq. (S5.3) (sign)	`prop_expected_captivity_gamma`, `prop_expected_captivity_sgn`	Part.
Proposition S6.1	`prop_optimal_gamma_beliefs`	Hyp.
Theorem S7.6	`conditionalImplementability`	Hyp.

Note. `[Proved]`: machine-checked in Lean 4. `[Part.]`: partial certificate (sign of (S5.3) given the analytic chain). `[Hyp.]`: declared interface; proved analytically in this supplement or cited literature. *Source:* Analytical proofs in this supplement; Lean identifiers in `lean/Appendix3Proofs.lean`. Software: Lean 4 (`lean-toolchain`); Mathlib via Lake.

3. Section S9 and Table S9.1 document which results are `[Proved]` vs. `[Hyp.]`.
4. (Optional) Compile `tex/Appendix_3.pdf` from `tex/` with `pdflatex` and `bibtex`.

Section S9 (Table S9.1) documents which results are machine-checked vs. hypothetical. All paths are relative to `Appendix_3/`; see `README.txt` for the full build workflow.